CLINICAL GENOMICS & PHARMACOGENOMICS ACTION BRIEF

Patient: Daniel Ehrle (Daniel_Ehrle) · **Date:** October 13, 2026 · **Primary Audience:** Patient, Primary Care Physician (PCP) & Specialists

SECTION 1 · PRIORITY GENOMIC FINDINGS, CARRIER STATUS & CLINICAL MEANING

GENE & BIOLOGICAL SYSTEM	VARIANT & ZYGOSITY	CLASSIFICATION & CONDITION	CLINICAL IMPACT & PRACTICAL GUIDANCE
CBLIF **CBLIF** Multisystem	**c.79+1G>A · rs147785187** **Heterozygous** Autosomal Recessive	**Pathogenic** Intrinsic Factor Deficiency / Cobalamin Malabsorption	Single-copy heterozygous carrier preserves intrinsic factor production. Check periodic serum B12 during routine wellness visits.
GJB2 **GJB2** Multisystem	**p.Met34Thr · c.101T>C · rs35887622** **Heterozygous** Autosomal Recessive, Autosomal Dominant	**Pathogenic** Autosomal Recessive Deafness 1A / Dominant 3A	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
VDR **VDR** Multisystem	**c.-1172A>G · rs4516035** **Homozygous** Pathway Modifier	**Likely pathogenic** Chronic obstructive pulmonary disease	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
F5 **F5** Multisystem	**p.Arg534Gln · c.1601G>A · rs6025** **Heterozygous** Autosomal recessive / AD	**drug response** Congenital factor V deficiency	Avoid systemic oral estrogen therapy (thromboembolism risk). Implement standard perioperative clot prophylaxis during major surgery.
PEX6 **PEX6** Heart & Cardiovascular	**p.Val788Met · c.2362G>A · rs267608240** **Heterozygous** Autosomal recessive	**Conflicting classifications of pathogenicity** Peroxisome biogenesis disorder 4A (Zellweger)	Single-copy heterozygous carrier preserves normal peroxisomal function. Standard annual liver and lipid wellness panels.
GJB3 **GJB3** Heart & Cardiovascular	**p.Arg75Cys · c.223C>T · rs370476720** **Heterozygous** Autosomal recessive	**Conflicting classifications of pathogenicity** Inborn genetic diseases	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
SCN4A **SCN4A** Heart & Cardiovascular	**p.His599Arg · c.1796A>G · rs187401185** **Heterozygous** Autosomal dominant / AR	**Conflicting classifications of pathogenicity** Paramyotonia congenita of Von Eulenburg	Maintain normal electrolyte balance. Inform anesthesiology regarding depolarizing muscle relaxants prior to general surgery.

PRESCRIBING SAFETY & PREVENTIVE CARE SCHEDULE

Patient: Daniel Ehrle (Daniel_Ehrle) · High-Priority Prescribing Safety & Annual Checkup Topics

SECTION 2 · HIGH-PRIORITY MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA)

GENE & VARIANT / DIPLOTYPE	PRESCRIBING ACTION	AFFECTED MEDICATION(S)	PRESCRIBING RULE & CLINICAL GUIDANCE
F5 · p.Arg534Gln · c.1601G>A · rs6025 (Heterozygous APC Resistance)	🚨 LIFE-THREATENING RISK · AVOID ESTROGENS	**Systemic Estrogens, Combined Oral Contraceptives, HRT**	**CRITICAL HAZARD:** Confers Activated Protein C (APC) resistance with 35-fold amplified risk of venous thromboembolism (VTE) and pulmonary embolism with systemic estrogen exposure. Avoid oral estrogen formulations; implement situational perioperative VTE prophylaxis.
CYP2C9 · *1/*2 [R144C (rs1799853) C>T] (Intermediate Metabolizer)	⬇️ HIGH RISK · REDUCE DOSE	**Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin)**	Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and monitor for GI/cardiovascular adverse effects. Initiate at reduced starting dose per CPIC.
CYP2C19 · *1/*17 [I331V (rs3758581) A>G, expression (rs12248560) C>T] (Rapid Metabolizer)	⬇️ HIGH RISK · REDUCE DOSE	**Voriconazole**	Increased clearance and lower trough concentrations. Perform therapeutic drug monitoring (TDM) to avoid subtherapeutic levels. Initiate at reduced starting dose per CPIC.
CYP3A5 · *3/*3 [splice defect (rs776746) T>C] (Poor Metabolizer)	⬇️ HIGH RISK · REDUCE DOSE	**Ondansetron**	Genotype CC is associated with decreased metabolism of ondansetron as compared to genotypes CT + TT. Initiate at reduced starting dose per CPIC.
UGT1A1 · *1/*28_80 [5'UTR (rs887829) C>T, Promoter (rs3064744) insTA] (Intermediate Metabolizer)	⚠️ USE WITH CAUTION	**Belinostat, Irinotecan (Camptosar)**	Monitor for increased hematologic toxicity. Monitor clinical response and consider dose adjustment per CPIC.

SECTION 3 · HIGH-LEVEL PREVENTIVE CARE & ANNUAL CHECKUP DISCUSSION POINTS

GENOMIC FINDING	BIOLOGICAL PATHWAY	PREVENTIVE CARE TOPIC	CARE SETTING & TIMING	CLINICAL CONTEXT & GUIDANCE
CBLIF c.79+1G>A · rs147785187 Heterozygous (Autosomal Recessive)	**CBLIF · Vitamin B12 (Cobalamin) & Folate Absorption** Pathogenic	**Periodic Vitamin B12 & Folate Check** Routine Wellness Lab Panel	**Primary Care Physician (PCP)** Routine Annual or Biennial Checkup	Single-copy heterozygous carrier preserves intrinsic factor production. Check periodic serum B12 during routine wellness visits.
GJB2 p.Met34Thr · c.101T>C · rs35887622 Heterozygous (Autosomal Recessive, Autosomal Dominant)	**GJB2 · Cochlear Gap Junction & Hearing Health** Pathogenic	**Baseline Hearing Check & Ototoxic Medication Precaution** Routine Preventive Care	**Primary Care Physician (PCP)** & **Audiology** Baseline check; repeat if symptoms arise	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
VDR c.-1172A>G · rs4516035 Homozygous (Pathway Modifier)	**VDR · Vitamin D Signaling & Bone Health** Likely pathogenic	**Routine Vitamin D & Calcium Check** Annual Wellness Panel	**Primary Care Physician (PCP)** Annual PCP Checkup	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
F5 p.Arg534Gln · c.1601G>A · rs6025 Heterozygous (Autosomal recessive / AD)	**F5 · Blood Coagulation & Clot Risk Management** drug response	**Situational Clot Prevention (Surgery, Immobility, Estrogen Avoidance)** Perioperative & Prescribing Precaution	**Primary Care Physician (PCP)** Situational (Before surgery, long travel, or hormone therapy)	Avoid systemic oral estrogen therapy (thromboembolism risk). Implement standard perioperative clot prophylaxis during major surgery.
PEX6 p.Val788Met · c.2362G>A · rs267608240 Heterozygous (Autosomal recessive)	**PEX6 · Peroxisomal & Lipid Metabolism** Conflicting classifications of pathogenicity	**Routine Liver & Lipid Wellness Panel** Standard Annual Physical Bloodwork	**Primary Care Physician (PCP)** Annual PCP Checkup	Single-copy heterozygous carrier preserves normal peroxisomal function. Standard annual liver and lipid wellness panels.

PHARMACOGENOMIC REFERENCE & CLINICAL PERSPECTIVE

Patient: Daniel Ehrle (Daniel_Ehrle) · Actionable Pharmacogenomic Guide & Physician Handoff Note

SECTION 4 · ACTIONABLE PHARMACOGENOMIC PRESCRIBING GUIDE (CPIC / DPWG)

PHARMACOGENE & DIPLOTYPE	METABOLIZER PHENOTYPE	ACTIONABLE MEDICATION SUBSTRATES	CPIC CLINICAL DOSING GUIDANCE
CYP2C19 *1/*17 I331V (rs3758581) A>G, expression (rs12248560) C>T	**Rapid Metabolizer**	**Reduce Dose: Voriconazole** **Increase Dose:** Omeprazole / Pantoprazole / Lansoprazole **Caution/Monitor:** Citalopram (Celexa), Escitalopram (Lexapro), Sertraline (Zoloft)	Increased clearance and lower trough concentrations. Perform therapeutic drug monitoring (TDM) to avoid subtherapeutic levels.
CYP2C9 *1/*2 R144C (rs1799853) C>T	**Intermediate Metabolizer**	**Reduce Dose: Celecoxib (Celebrex)**, **Phenytoin (Dilantin)**, **Warfarin (Coumadin)** **Caution/Monitor:** Flurbiprofen / Ibuprofen	Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and monitor for GI/cardiovascular adverse effects.
CYP3A5 *3/*3 splice defect (rs776746) T>C	**Poor Metabolizer**	**Reduce Dose: Ondansetron** **Increase Dose:** Vardenafil **Caution/Monitor:** 4-Hydroxycyclophosphamide, Atazanavir / Ritonavir	Genotype CC is associated with decreased metabolism of ondansetron as compared to genotypes CT + TT.
UGT1A1 *1/*28_80 5'UTR (rs887829) C>T, Promoter (rs3064744) insTA	**Intermediate Metabolizer**	**Caution/Monitor:** Belinostat, Irinotecan (Camptosar), Atazanavir (Reyataz)	Monitor for increased hematologic toxicity.
F5 rs6025 Factor V Leiden (T)/rs6025 reference (C) p.Thr295Ala (rs371760153) [Heterozygous]	**Activated Protein C (APC) Resistance**	**Avoid: Combined Oral Contraceptives (Estrogen/Progestin)**, **Hormone Replacement Therapy (Estrogen Hrt)**	Factor V Leiden carrier (rs6025 C>T heterozygous). 3- to 8-fold increased baseline venous thromboembolism (VTE) risk, amplified up to 35-fold with …

SECTION 5 · CLINICAL PERSPECTIVE: BALANCING PROACTIVE CARE VS. OVER-INTERVENTION

✅ Why These Findings Are Useful

- **Proactive Medication Safety:** Recording pharmacogenomic metabolizer status in the medical chart helps select effective medications and avoid preventable side effects before a prescription is started.
- **Informed Annual Checkups:** Highlighting relevant biological pathways allows the Primary Care Physician to tailor routine wellness checks and age-appropriate preventive screening.

🛑 Why Aggressive Intervention Is Not Needed

- **Single-Copy Carrier Reassurance:** No two-copy (biallelic) recessive disease genotypes are present; single-copy recessive carriers retain a functional gene copy and maintain normal baseline health.
- **Modifier Variants Are Not Diagnoses:** Common pathway modifiers indicate mild physiological tendencies and should never trigger unnecessary procedures or unprovoked medication changes.

SECTION 6 · COPY-PASTE PHYSICIAN HANDOFF / EHR CHART SUMMARY

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CLINICAL GENOMICS & PHARMACOGENOMICS SUMMARY (Daniel Ehrle | ID: Daniel_Ehrle)
• PRIORITY GENOMIC & CARRIER FINDINGS: CBLIF (c.79+1G>A · rs147785187, Heterozygous), GJB2 (p.Met34Thr · c.101T>C · rs35887622, Heterozygous), VDR (c.-1172A>G · rs4516035, Homozygous), F5 (p.Arg534Gln · c.1601G>A · rs6025, Heterozygous), PEX6 (p.Val788Met · c.2362G>A · rs267608240, Heterozygous), GJB3 (p.Arg75Cys · c.223C>T · rs370476720, Heterozygous). Single-copy (heterozygous) recessive carrier states retain one functional gene copy and do not cause primary recessive disease.
• MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA):
  - AVOID / CONTRAINDICATED: Systemic Estrogens, Combined Oral Contraceptives, HRT
  - REDUCE STARTING DOSE: Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin) · Voriconazole · Ondansetron
  - RAPID METABOLISM / MONITOR: Belinostat, Irinotecan (Camptosar) · Isoniazid
• PREVENTIVE CARE PLAN: Review high-level wellness and screening topics in Section 3 during routine primary care checkups.
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